attention to these attributes while designing and developing Dapps. This chapter will provide a high-level view of these attributes, challenges, existing solutions, potential opportunities, and the road ahead.

12.1 Decentralized identity

What is your name? How are you identified? Identity is a fundamental requirement for interacting with any system, computing or noncomputing. You need an instrument such as a driver's license to identify you for many of your routine activities, such as cashing a large check or taking an airline flight. You use a student ID card to avail yourself of the services of a university. But these identities are issued by a central authority after verification of your credentials, such as your Social Security number. For decentralized applications, there is no central authority for assigning identity for the participants. A decentralized system is made up of unknown participants who are potentially from anywhere in the world. In such a system, the challenge is how to

- Define a unique identity for participants
- Create and assign it to the participants
- Make it unique for every participant
- Manage (restore and remember) it

In addressing these concerns, blockchains rely on two fundamental concepts: cryptographic algorithms and larger address space (256 bits versus 64 bits). As you'll recall from chapter 5, the Ethereum identity of a participant is 160 bits. It is derived from a private-public key pair of 256 bits, using hashing algorithms for uniqueness, and can be self-generated. This kind